

The main objective of this task was to develop a **Text Summarization (Extractive)** chatbot capable of understanding user intents, accepting long input text, and producing summarized output using an extractive approach.

The goal was also to ensure the chatbot could handle various queries (greetings, summary requests, and exit commands), while performing text preprocessing, intent matching, and response generation seamlessly.

Implementation Summary

During implementation, three code versions were developed and tested step by step. Each version was preserved in the Colab notebook for record-keeping, showing how the chatbot evolved from an error-prone build to a fully functional version.

Version 1 – Using `gensim.summarization` (TextRank)

- **Action:** Initially implemented extractive summarization using the `gensim.summarization.summarize()` function.
- **Issue:** Execution failed with
- `ModuleNotFoundError: No module named 'numpy.rec'`

This happened because `gensim`'s summarization module is deprecated and incompatible with the latest NumPy (2.0+) used in Google Colab.

- **Result:** No summary was generated; the notebook cell produced an error.
- **Decision:** This version was **not deleted** but retained for documentation purposes.

Version 2 – Using `sumy` (LexRank)

- **Action:** Replaced `gensim` with the `sumy` library, which provides stable and modern extractive summarization algorithms (LexRank).
- **Outcome:** The library installed successfully, and the chatbot executed without errors.
- **Observation:**
 - The bot responded correctly to greetings and exit commands.

- However, when a paragraph was entered for summarization, it **did not generate a summary** and instead gave a greeting-like response.
- **Reason:**
 - The intent-matching logic only recognized summarization when explicit keywords like “summarize” or “shorten” were present.
 - Long text inputs were not automatically identified as summarization tasks.
- **Result:** Partial success — code ran without crashes but didn’t summarize text properly.
- **Decision:** This version was also **kept** in the notebook since it produced partial output and validated the summarizer installation.

Version 3 – Final Working Model

- **Action:**
 - Retained the sumy LexRank summarizer.
 - Improved **intent detection** to automatically identify long paragraphs (≥40 words) as summarization inputs.
 - Installed both `nltk.download('punkt')` and `nltk.download('punkt_tab')` to fix tokenizer dependency issues.
 - Added robust error handling and output formatting.
- **Result:**
 - Bot successfully generated extractive summaries.
 - Greeted and exited appropriately.
 - Responded intelligently to long text without needing explicit summarization keywords.
 - Saved final script as `Text_Summarization_Chatbot_Working.py`.

Sample Output:

User: Artificial Intelligence (AI) is a rapidly advancing field...

Bot: Summary:

Artificial Intelligence (AI) is a rapidly advancing field.

AI applications are seen in healthcare, finance, transportation, and entertainment.

It raises both opportunities and ethical challenges that must be addressed.

Why I Chose This Model

I selected the **Extractive Summarization approach** using the **LexRank algorithm (Sumy)** because:

- It identifies the most important sentences directly from the text instead of generating new ones, ensuring factual accuracy.
- LexRank is **unsupervised**, requiring no training data — ideal for lightweight chatbot applications.
- It performs well for **general-purpose text summarization** without complex neural model dependencies.
- The sumy library is stable and compatible with the latest Python and Colab environments, unlike older gensim.summarization.

Detailed Step Explanation

Step	Description
1. Define Intents	Created greet, summarize, and bye intents with example queries.
2. Preprocess Input Text	Cleaned user input using regex to remove unwanted characters and normalize spaces.
3. Match Input to Intents	Used regex pattern matching; later added logic to auto-detect long paragraphs as summary requests.
4. Generate Responses	Implemented extractive summarization using LexRank Summarizer from the sumy library.
5. Test Chatbot	Tested with multiple user inputs — greetings, short queries, long paragraphs, and exit messages.
6. Save Script	Saved the final chatbot as Text_Summarization_Chatbot_Working.py for documentation and reuse.

Conclusion

The project was successfully completed.

Three code versions were tested sequentially:

- The first version failed due to library incompatibility.
- The second produced partial responses without errors.
- The third version met all requirements and generated accurate extractive summaries.

This progression demonstrates clear debugging, model improvement, and logical refinement.

The final chatbot is efficient, accurate, and ready for presentation or deployment in future text-processing pipelines.